

1. Data Link Layer (DLL) Foundations & Framing

Module 01

- **DLL Core Responsibilities:** Framing, Flow Control, Error Control, Physical Addressing (MAC), and Channel Access Control (MAC Sublayer).
- **LLC (802.2) vs MAC:** LLC handles upper protocol multiplexing & flow control (medium independent). MAC handles channel access, framing & contention.
- **Framing Methods:**
 - *Character Count:* Frame header carries total byte count. Vulnerable: 1 corrupted count bit desynchronizes all subsequent frames!
 - *Byte / Character Stuffing:* Flag byte ('0x7E') delimits frame. If Flag occurs in payload, prefix with 'ESC' ('0x1B'). If ESC in payload, prefix with 'ESC'.
 - *Bit Stuffing (5-Ones Rule):* Flag = '01111110'. Sender automatically stuffs a '0' after five consecutive '1's ('111110'). Receiver unstuffs it.

Bit Stuffing: 011111101111110 -> 011111(0)1011111(0)10

2. Pure ALOHA vs Slotted ALOHA

Module 02

- **Pure ALOHA (Continuous):** Station transmits immediately whenever data is ready.
 - Vulnerable Time: $T_{vu1n} = 2 \cdot T_{fr}$
 - Poisson Throughput: $S = G \cdot e^{(-2G)}$
 - Max Throughput: $S_{max} = 1 / (2e) \approx 18.4\%$ at $G = 0.5$.
- **Slotted ALOHA (Discrete Clock Slots):** Station can only transmit at beginning of slot (Slot duration = T_{fr}).
 - Vulnerable Time: $T_{vu1n} = 1 \cdot T_{fr}$ (Halved!)
 - Poisson Throughput: $S = G \cdot e^{(-G)}$
 - Max Throughput: $S_{max} = 1 / e \approx 36.8\%$ at $G = 1.0$ (Doubled!).

AKTU Poisson: $P(k) = (G^k \cdot e^{(-G)}) / k!$ | $P(0) = e^{(-G)}$ (1 slot)

3. Carrier Sense Multiple Access (CSMA) & Persistence Strategies

Module 03

1-Persistent CSMA:

- Continuously senses channel.
- If Idle: Transmits immediately ($p=1$).
- If Busy: Keeps continuous listening.
- High collision if 2 stations wait for busy channel to free!

Non-Persistent CSMA:

- Senses channel once.
- If Idle: Transmits immediately.
- If Busy: Waits random backoff time before sensing again.
- Reduces collisions; wastes channel idle time. Peak ~0.8.

p-Persistent CSMA:

- Used in slotted channels.
- If Idle: Transmits with prob p ; defers to next slot with prob $1 - p$.
- Optimal: $N \cdot p < 1$ to prevent collisions.

Collision Window: t in $[0, T_p]$ due to propagation delay. Carrier sensing cannot see ahead of wavefront!

4. CSMA with Collision Detection (CSMA/CD)

Module 04

- **Mechanism:** "Listen While Talk". Transceiver checks voltage: $V_{chan} > 2 \cdot V_{tx}$ indicates collision.
- **Collision Action:** Immediately abort frame + Broadcast 32/48-bit **Jam Signal** + Increment k (collision count).
- **Binary Exponential Backoff (BEB):** Wait $R \cdot Slot_Time$ where $0 \leq R \leq 2^k - 1$ (Max $k=10$; Abort at $k=16$).
- **Core Derivation (64-Byte Frame):**

$T_t \geq 2 \cdot T_p \implies L_{min} = 2 \cdot T_p \cdot Bandwidth = 512 \text{ bits} = 64 \text{ Bytes}$

5. CSMA with Collision Avoidance (CSMA/CA)

Module 05

- **Why CSMA/CD Fails in Wireless:**
 - Dynamic Range: Transmit power drowns received power (Tx blinds Rx).
 - Hidden Station Problem: A & C cannot hear each other; collide at B.
 - Exposed Station: C refrains from sending to D due to hearing A-B.
- **Interframe Spaces (IFS):** $SIFS < PIFS < DIFS < EIFS$ (Shortest IFS = Highest Priority; SIFS for ACK/CTS).
- **4-Way RTS/CTS Handshake:** RTS carries duration; CTS echoes it. Stations overhearing CTS set **NAV** countdown timer (Virtual Sensing).

$NAV = SIFS + T_{CTS} + SIFS + T_{DATA} + SIFS + T_{ACK}$

6. Controlled Access & CDMA Channelization

Module 06

Controlled Access Methods:

- **Reservation:** N bit reservation interval precedes N data frames.
- **Polling:** Primary-Secondary architecture. Poll (asks data), Select (sends data). Single point of failure at Primary.
- **Token Passing:** Special Token frame circulating in logical ring. Station holding token transmits for Token Holding Time (THT). Zero collisions!

CDMA & Orthogonal Walsh Codes:

- Walsh Recursive Construction: $W_{2N} = [W_N, W_N; W_N, -W_N]$
- Properties: $W_i \cdot W_i = N$ and $W_i \cdot W_j = 0$ ($i \neq j$)
- Data Encoding: Bit 0 = -1, Bit 1 = +1, Silent = 0. Transmitted: $S_i = d_i \cdot W_i$.
- Channel Signal: $C = \sum S_i$ | Receiver Decoding: $d_i = (C \cdot W_i) / N$

Example: $C = [1, 1, -1, 3], W2 = [1, -1, 1, -1] \rightarrow C \cdot W2 = 1 - 1 - 1 - 3 = -4 \rightarrow d2 = -4/4 = -1$ (Bit 0 Decoded!)

7. Flow Control Foundations & Stop-and-Wait ARQ Protocol

Module 07

- **Flow Control:** Prevents fast sender from overflowing slow receiver buffer.
- **Error Control (ARQ):** Adds Timer + Retransmission + Sequence Numbers.
- **1-Bit Sequence Numbering (0 & 1):** Prevents duplicate frame acceptance when ACK is lost! Receiver ACK n expects frame n.
- **Piggybacking:** Outgoing ACK combined inside outgoing data frame header.

Total Time $T_{total} = T_t + 2 \cdot T_p$ (Neglecting Tack, Tproc)
Parameter $a = T_p / T_t = (d / v) / (L / \text{Bandwidth})$
Efficiency $\eta = T_t / (T_t + 2 \cdot T_p) = 1 / (1 + 2a)$ | Throughput $= \eta \cdot B$
Physical Impact of 'a': If $a \gg 1$ (Satellite link, $a=250$): η drops to $1 / (1 + 500) = 0.2\%$! Motivates Sliding Window Pipelining.

8. Sliding Window Protocols: Go-Back-N vs Selective Repeat ARQ

Module 08 & 09

Parameter / Feature	Stop-and-Wait ARQ	Go-Back-N (GBN) ARQ	Selective Repeat (SR) ARQ
Sender Window (Ws)	1	$2^m - 1$	$2^{(m-1)}$
Receiver Window (Wr)	1	1 (Strictly sequential)	$2^{(m-1)}$ (Out-of-order buffer!)
Window Sizing Proof	$Ws + Wr \leq 2^m \rightarrow 1 + 1 \leq 2$	$Ws + 1 \leq 2^m \implies Ws \leq 2^m - 1$	$Ws + Wr \leq 2^m \implies 2 \cdot Ws \leq 2^m \implies Ws \leq 2^{(m-1)}$
Out-of-Order Buffering	No	No (Subsequent frames discarded!)	Yes (Stored in reassembly buffer)
Acknowledgment Type	Individual ACK	Cumulative ACK (ACK n confirms up to n-1)	Individual ACK + NAK (Selective Reject)
Retransmission Scope	1 frame	All N frames in window (Go-Back penalty)	Only the damaged / lost frame
Number of Timers	1 timer	1 timer (For oldest unacknowledged frame Sf)	Multiple independent timers (1 per frame)
Efficiency (η)	$1 / (1 + 2a)$	$Ws / (1 + 2a)$ (for $Ws < 1+2a$; 100% if $Ws \geq 1+2a$)	$Ws / (1 + 2a)$ (High utilization even on lossy link)

GBN 100% Efficiency Condition: $Ws \geq 1 + 2a \implies m \geq \lceil \log_2(2 + 2a) \rceil$
Why $Ws \leq 2^m - 1$? If $Ws = 2^m$ and all ACKs lost, sender retransmits old window; receiver cannot distinguish old duplicate from new frame!

9. Error Detection: Parity, Checksum & CRC

Module 10

- Error Types:** Single-Bit vs Burst Error (Length = first to last flipped bit index).
- Parity:** 1D parity fails on even-bit flips. 2D Parity detects all 1, 2, 3-bit errors; corrects 1-bit error via row-column intersection.
- Internet Checksum:** 16-bit 1's complement sum. Wrap-around carry added to LSB. Final result complemented. Receiver sum must be all 1s.
- Cyclic Redundancy Check (CRC):** Modulo-2 XOR division.
 - Degree of $G(x)$ is r . Append r zeros to data.
 - Divide by $G(x)$ using Modulo-2 arithmetic.
 - Remainder R is CRC (FCS). Codeword = Data + R .
 - Receiver checks if $\text{Codeword} \bmod G(x) == 0$.

CRC-32: Used in 802.3 Ethernet FCS. Detects 100% burst errors ≤ 32 bits!

10. Error Correction & Hamming Code

Module 11

- Hamming Distance ($d_{\min}$):**
 - To detect d errors: $d_{\min} \geq d + 1$
 - To correct t errors: $d_{\min} \geq 2t + 1$
- Redundancy Formula:** $2^r \geq m + r + 1$ (For $m=4$ bits data $\rightarrow r=3$ bits $\rightarrow$ Hamming (7, 4) Code).
- Bit Positions:** Parity at powers of 2 (1, 2, 4). Data at 3, 5, 6, 7.

$p1 = d1 \oplus d2 \oplus d4 \mid p2 = d1 \oplus d3 \oplus d4 \mid p4 = d2 \oplus d3 \oplus d4$
- Syndrome Word ($S4\ S2\ S1$):**

$S1 = b1 \oplus b3 \oplus b5 \oplus b7 \mid S2 = b2 \oplus b3 \oplus b6 \oplus b7 \mid S4 = b4 \oplus b5 \oplus b6 \oplus b7$

Value of ($S4\ S2\ S1$) directly gives the erroneous bit index! Invert to correct.

11. IEEE 802.3 Ethernet Frame & Standards Evolution

Module 12

Preamble	SFD	Dest MAC	Src MAC	Length/Type	Data Payload (LLC PDU)	FCS (CRC-32)
7 Bytes (1010..)	1 Byte (10101011)	6 Bytes (48b)	6 Bytes (48b)	2 Bytes (Type/Len)	46 to 1500 Bytes (Padding if <46B)	4 Bytes

- Frame Limits:** Minimum = 64 Bytes (512 bits); Maximum = 1518 Bytes.
- MAC Address (48 bits):** First 24 bits = OUI (Vendor ID); Last 24 bits = NIC serial. Broadcast = `FF:FF:FF:FF:FF:FF`.
- Ethernet Family:** 10Base5 (Thick), 10Base2 (Thin), 10Base-T (Twisted pair). Fast Ethernet (100 Mbps, 802.3u). Gigabit Ethernet (1 Gbps, 802.3z). 10G Ethernet (802.3ae, Full-Duplex only, No CSMA/CD).

12. Token Ring (802.5) & FDDI

Module 13

- IEEE 802.5 Token Ring:** Physical star, logical ring via MSAU. Token = SD (1B) + AC (1B) + ED (1B). Active Monitor clears orphan frames & lost tokens. Early Token Release in 16 Mbps.
- FDDI (100 Mbps Optical Fiber):**
 - Dual Counter-Rotating Ring (Primary CW + Secondary CCW).
 - Self-Healing: Cable cut triggers wrap-around into 1 single ring!
 - Timed Token Protocol: $THT = \max(0, TTRT - TRT)$.

13. Bridges, Switches & STP (802.1D)

Module 14

- Collision vs Broadcast Domains:** Switch splits collision domains (1 per port); preserves 1 broadcast domain.
- Transparent Bridge Learning:** Learn (Source MAC), Forward/Filter (Dest MAC), Flood (Unknown/Broadcast).
- Spanning Tree Protocol (STP 802.1D):** Prevents broadcast storms & loops.
 - 1. Elect Root Bridge (Lowest Bridge ID: Priority + MAC).
 - 2. Select Root Port (RP) per non-root bridge (Lowest path cost).
 - 3. Select Designated Port (DP) per segment.
 - 4. Place redundant ports in **BLOCKING** state!